

COL RIVER BAR BUOY 46029, USA

WMO: 992370

Lat: 46.143N

Lon: 124.485W

Elev: 0

StdP: 14.70

Time Zone: -8.00 (W08)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	35.9	39.0	23.5	17.8	38.1	28.0	21.9	41.1	38.6	48.2	35.0	49.4	15.7	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.1	63.9	61.8	63.1	61.1	62.3	60.6	62.3	62.4	61.4	61.6	60.6	60.9	10.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	83.9	62.3	61.3	81.1	61.4	60.4	78.6	60.7	27.9	62.0	27.2	61.7	26.8	61.1	65.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.1	31.6	26.5	DB	32.6	65.3	3.1	1.3	30.4	66.2	28.6	67.0	26.9	67.7	24.7	68.7	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	47.8	46.7	47.3	48.6	51.9	55.7	58.7	59.1	58.2	55.2	50.8	47.7
	DBStd	5.45	3.48	3.63	2.82	2.65	2.74	2.27	2.31	2.02	2.21	2.82	3.33	3.86
	HDD50	458	83	99	92	55	12	0	0	0	0	2	28	87
	HDD65	4624	535	512	549	491	406	279	196	183	204	305	427	537
	CDD50	1310	13	7	7	14	71	171	270	282	246	163	51	15
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	13.9	16.4	15.3	15.2	14.0	12.0	12.1	11.7	10.6	11.8	13.8	16.2	17.8
Precipitation	PrecAvg	74.1	11.3	7.7	7.9	5.9	3.1	2.8	1.1	1.8	1.9	7.5	11.2	11.6
	PrecMax	97.2	23.1	21.3	17.0	11.8	5.2	5.2	3.3	5.2	7.6	15.1	21.3	20.3
	PrecMin	55.5	4.4	0.7	0.8	2.3	0.4	0.7	0.2	0.1	0.0	1.7	3.2	5.4
	PrecStd	13.8	4.4	4.9	3.6	2.6	1.3	1.2	0.7	1.5	1.8	3.5	5.3	4.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	55.6	54.9	58.3	61.6	62.4	64.8	64.5	65.0	63.0	59.0	57.5
		MCWB	50.5	48.2	50.0	52.9	57.9	59.5	62.1	62.8	64.3	55.4	55.3	57.5
	2%	DB	54.2	54.1	53.7	55.0	58.3	60.6	63.6	63.7	63.4	61.4	57.3	55.2
		MCWB	50.8	48.3	50.2	51.6	55.4	59.4	62.1	62.1	60.6	56.4	54.5	53.6
	5%	DB	53.0	52.2	52.6	53.2	56.8	59.7	62.8	62.8	62.5	60.2	56.2	53.6
		MCWB	50.7	48.4	49.9	50.6	54.3	58.7	61.5	61.3	60.1	57.1	54.0	52.0
	10%	DB	52.0	50.8	51.1	52.1	55.5	59.0	62.1	62.1	61.7	59.1	55.1	52.4
		MCWB	50.0	48.1	49.1	50.2	53.5	57.7	60.9	60.6	59.7	57.0	53.3	50.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.3	51.0	52.3	53.7	59.2	61.2	63.2	64.1	64.0	59.3	56.3	54.3
		MCDB	53.4	51.2	52.7	56.6	61.3	61.3	63.2	64.1	64.0	59.5	56.5	54.7
	2%	WB	52.2	50.1	50.9	52.2	56.9	60.2	62.1	62.4	61.5	58.2	55.4	52.3
		MCDB	52.6	50.6	51.4	53.5	58.7	60.3	62.1	62.5	61.6	58.5	55.6	52.7
	5%	WB	51.4	49.3	50.1	51.4	55.3	59.3	61.2	61.5	60.3	57.3	54.7	51.1
		MCDB	51.9	50.0	50.8	52.0	56.5	59.4	61.3	61.7	60.7	57.8	55.0	51.5
	10%	WB	50.4	48.6	49.3	50.4	54.3	57.7	60.4	60.8	59.5	56.5	53.6	50.0
		MCDB	51.3	49.5	50.2	51.2	55.6	58.2	60.6	61.2	60.0	57.2	54.1	50.8
Mean Daily Temperature Range	5% DB	MDBR	3.8	3.7	3.7	3.4	2.8	2.7	2.7	3.1	3.3	3.5	4.0	4.1
		MCDBR	4.0	4.1	4.5	4.3	3.8	3.6	3.1	3.7	3.9	3.9	4.1	4.6
		MCWBR	3.8	3.6	4.1	4.7	3.6	3.3	3.4	3.7	3.9	4.0	3.9	5.5
	5% WB	MCDBR	3.5	3.3	3.7	3.6	3.7	3.1	3.3	3.5	3.9	3.5	4.0	4.1
MCWBR		4.0	4.0	3.9	4.0	3.3	3.5	3.1	3.6	4.0	4.3	4.3	5.4	
Clear-Sky Solar Irradiance	taub	0.327	0.324	0.340	0.374	0.390	0.363	0.352	0.354	0.349	0.341	0.330	0.322	
	taud	2.510	2.552	2.523	2.392	2.371	2.478	2.536	2.536	2.547	2.552	2.523	2.500	
	Ebn at Noon	246	273	283	280	279	286	288	284	276	262	243	233	
	Edh at Noon	21	24	29	35	37	34	31	30	28	24	21	19	
All-Sky Solar Radiation	RadAvg	370	649	1006	1437	1706	1841	1810	1571	1303	801	440	313	
	RadStd	46	69	97	112	142	119	160	140	97	78	51	38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+32

Nomenclature: See separate page